

CareerCompass

An ML-Based System for Tech Career Trend Prediction and Skill Recommendation

Project Report

Machine Learning — Course Project

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1. System Ideation

1.1 Problem Statement

Final-year and pre-final-year engineering students routinely make two high-stakes decisions — which skills to invest time in, and which companies to target — using fragmented, anecdotal, and often outdated information: senior placement stories, scattered LinkedIn posts, and generic “top skills to learn” listicles. None of these sources are grounded in a structured, continuously updated dataset that reflects the actual state of hiring demand, compensation, and workforce contraction in the tech industry at a given point in time.

This creates a recurring mismatch: students frequently prepare for roles or companies whose hiring patterns have already shifted, or invest in skills whose market demand is flat or declining, simply because no accessible tool synthesizes postings data, compensation data, and layoff data into a single, personalized, evidence-backed recommendation.

1.2 Proposed System

CareerCompass is proposed as a machine-learning-driven advisory system that accepts three inputs from a student — their current skill set, a target (“dream”) company, and a salary expectation — and returns a structured, evidence-grounded output covering four elements:

- Market-trend standing of the student’s target company (growing / stable / contracting hiring activity, based on recent posting volume and layoff signals).
- A ranked shortlist of comparable or better-fit companies, scored by skill match, compensation alignment, and hiring-trend health.
- A predicted realistic salary band for the student’s skill set and target role, benchmarked against their stated expectation.
- A prioritized skill-gap list — skills to acquire, ranked by demand trend and frequency in postings for the student’s target roles.

The system is explicitly positioned as a decision-support tool grounded in a purpose-built dataset, not a general-purpose chatbot. Every recommendation it produces is traceable to a specific data point or model output — a posting count, a regression coefficient, a trend slope — which distinguishes it from LLM-based career advice that cannot be interrogated for its evidentiary basis in the same way.

1.3 Motivation and Related Work

Career-guidance and AI-coaching tools already exist commercially, and the space has grown substantially — the AI career-coaching market was valued near \$6.69 billion in 2026 and is projected to nearly double in four years, reflecting demand for guidance that keeps pace with a job market being restructured at the task level. Section 3 of this report surveys the closest existing systems in detail and identifies specific gaps this project addresses: none of the surveyed tools combine dream-company-specific benchmarking, salary-expectation reframing against real data, and explicit skill-drop guidance within a single transparent, academically reproducible system.

1.4 Scope Boundary: What Is ML vs. What Is Data Engineering

A deliberate design decision in this project is to be explicit about which components are genuinely learned models and which are deterministic aggregation or ranking logic. This distinction is documented up front rather than blurred, both for academic honesty and because it clarifies where the system's real predictive value lies.

Component	Category	Justification
Salary prediction	Machine Learning (regression)	Learned function of skills, role, location, seniority — genuinely benefits from a trained model over hand-written rules.
Skill demand trend classification	Machine Learning (time-series / lightweight classifier)	Trend direction is inferred from historical posting patterns rather than fixed thresholds.
Company recommendation ranking	Hybrid — similarity scoring (ML) + rule-based filtering	Vector similarity is a learned representation; salary/location filters are deterministic business rules layered on top.
Market-trend summary for a named company	Data aggregation / visualization	Time-series counts and layoff signals presented directly; no prediction is made about this specific company's future beyond descriptive trend.
Skill-gap list generation	Rule-based ranking (set difference + trend score)	A set-difference and sort operation; using a full model here would add complexity without added accuracy at this data scale.

2. Machine Learning Models

This section specifies, for each predictive task, the model family selected, the reasoning behind that choice, the input features, the target output, and the evaluation metric to be used. Model choices favor interpretable, tabular-data-appropriate methods over deep learning, both because the available dataset size does not justify deep architectures and because interpretability (e.g., feature importance) is directly useful for explaining recommendations to students and for defending design choices academically.

2.1 Salary Prediction Model

Task: Predict a realistic salary range for a student given their skills, target role, target company (or company tier), location, and seniority level.

Model: Gradient-boosted decision trees (XGBoost or LightGBM), with a linear regression baseline for comparison.

Gradient boosting is preferred over a neural network here because the input is predominantly tabular and mixed categorical/numeric data at a moderate dataset size (likely tens of thousands of rows), a regime where tree ensembles consistently outperform deep models and require far less tuning. Feature importance output from the trained model also gives a direct, presentable explanation for why a predicted salary came out the way it did — useful both for the student-facing UI and for the project defense.

Input Features

- One-hot / target-encoded role category and seniority level
- Multi-hot encoded skill vector (from the skills taxonomy, Section 3)
- Company size band and industry (categorical)
- Location (encoded; optionally cost-of-living adjusted)
- Posting recency (to capture time-based salary drift)

Target and Evaluation

- Target: salary_min and salary_max (two regression heads, or a single midpoint target with a separately modeled range width)
- Metrics: Mean Absolute Error (MAE) in currency units, and R^2 on a held-out test split; MAE is prioritized over RMSE for interpretability to a non-technical reader (“on average, predictions are off by ₹X”)

2.2 Skill Demand Trend Classifier

Task: Classify each skill in the taxonomy as Rising, Stable, or Declining based on its demand trajectory over the available historical window.

Model: Rolling-window linear regression slope per skill, thresholded into three classes; optionally compared against a simple logistic regression / decision tree classifier trained on engineered trend features for a stronger academic contribution.

This task is intentionally kept lightweight. A recurrent or transformer-based time-series model would be difficult to justify given a limited historical window (job-posting and survey datasets typically span a small number of years at monthly granularity) and would not meaningfully outperform a well-engineered slope-and-threshold approach or a shallow classifier. This choice is documented explicitly in Section 1.4 as a place where added model complexity would not be justified by the data available.

Input Features (per skill, per time window)

- Monthly posting count mentioning the skill (normalized by total postings that month, to separate real trend from overall market volume changes)
- Slope of a rolling linear fit over the trailing 6–12 months
- Year-over-year percentage change, where survey data (e.g. Stack Overflow Developer Survey) is available to extend the window

Target and Evaluation

- Target: 3-class label (Rising / Stable / Declining), assigned initially via slope thresholds and used as ground truth for a supervised classifier if one is trained
- Metrics: classification accuracy and per-class F1 score on a held-out set of skill-months; qualitative validation against known industry narratives (e.g. cloud/AI skills trending up) as a sanity check

2.3 Company Recommendation Engine

Task: Given a student's skill vector, target salary, and dream company, return a ranked list of companies that are a stronger or comparable fit.

Model: Content-based similarity scoring — cosine similarity between the student's skill vector and each company's aggregated hiring-skill vector — combined with rule-based filters for salary band and current hiring-trend health.

Collaborative filtering (the approach used by most commercial recommender systems) is deliberately not used here, since it requires a large volume of user-company interaction history that a college project's own dataset will not have. Content-based similarity against company hiring profiles built from the postings dataset is both feasible at this data scale and directly explainable: a recommendation can be shown alongside the specific skill overlap that produced it.

Input Features

- Student skill vector (multi-hot or TF-IDF weighted, from skills taxonomy)
- Company hiring-skill vector, aggregated from that company's postings over a trailing window
- Company hiring-trend score (from Section 2.2-style aggregation, applied at company level instead of skill level)
- Salary compatibility (student expectation vs. company's observed salary band for matching roles)

Target and Evaluation

- No single ground-truth label exists for “correct” recommendation; evaluation is via offline proxy metrics (e.g., does the model rank the student's actual dream company highly when skills strongly match its typical hiring profile) and qualitative review by the project team

2.4 Skill Gap Recommendation

Task: Given a student's current skills and their target role/company, output a ranked list of skills to acquire.

Model: Deterministic ranking — set difference between (skills frequently required for the target role at trend-healthy companies) and (student's current skills), ranked by a composite score of posting frequency and trend classification from Section 2.2.

This is intentionally not framed as a trained model. It is a ranking function over already-modeled quantities (skill trend class, posting frequency), and forcing a separate learned model onto this step would add complexity without a corresponding gain in accuracy or explainability — the composite score is already fully interpretable.

2.5 Model Pipeline Summary

Stage	Technique	Output consumed by
Data ingestion & cleaning	Pandas / regex-based normalization	All downstream stages
Skill extraction & taxonomy mapping	Rule-based matching + fuzzy matching (e.g. RapidFuzz)	Skill vectors used across 2.1–2.4
Salary prediction	XGBoost / LightGBM regression	Student-facing salary benchmark output
Skill trend classification	Rolling regression slope + optional shallow classifier	Skill-gap ranking, company scoring
Company similarity scoring	Cosine similarity + rule filters	Company recommendation list
Skill gap ranking	Weighted set-difference ranking	Final “skills to learn / drop” output

3. Dataset Schema

The system relies on a purpose-built relational dataset assembled from public sources (job-posting datasets, the Stack Overflow Developer Survey, and layoffs.fyi historical records). The schema below is designed to support all four modeling tasks in Section 2 without redundant duplication of skill or company information.

3.1 Entity-Relationship Overview

Six core tables and one logging table make up the schema: companies, job_postings, skills, posting_skills (a junction table), skill_trends (a derived/aggregated table), layoffs, and user_queries (application usage logging, not training data). Relationships are as follows: a company has many job_postings and many layoff records; a job_posting has many skills through posting_skills; a skill has many monthly entries in skill_trends.

3.2 Table: companies

Column	Type	Description
company_id	INTEGER (PK)	Unique identifier
name	VARCHAR(255)	Company name, normalized (e.g. “Google” not “Google LLC”)
industry	VARCHAR(100)	e.g. fintech, e-commerce, AI/ML, SaaS, core-tech
size_band	ENUM	startup / mid-size / large / enterprise-FAANG-tier
hq_location	VARCHAR(100)	Primary headquarters location
glassdoor_rating	FLOAT (nullable)	Optional enrichment field
created_at	TIMESTAMP	Record creation time

3.3 Table: job_postings

Column	Type	Description
posting_id	INTEGER (PK)	Unique identifier
company_id	INTEGER (FK)	References companies.company_id
role_title	VARCHAR(150)	Raw title as scraped/sourced
role_category	VARCHAR(50)	Normalized category: backend / frontend / ML / data / devops / mobile / other
seniority	ENUM	intern / junior / mid / senior / lead
salary_min	FLOAT (nullable)	Lower bound of stated salary range
salary_max	FLOAT (nullable)	Upper bound of stated salary range
location	VARCHAR(100)	Job location or “remote”
posted_date	DATE	Critical for all time-series trend analysis
source	VARCHAR(50)	Dataset/source identifier, for provenance

3.4 Table: skills

Column	Type	Description
skill_id	INTEGER (PK)	Unique identifier
skill_name	VARCHAR(100)	Normalized name (e.g. “React”, not “ReactJS”/“React.js” variants)
category	ENUM	language / framework / database / cloud-tool / soft-skill / other
aliases	TEXT	Comma-separated known synonyms, used for normalization matching

3.5 Table: posting_skills (junction)

Column	Type
posting_id	INTEGER (FK) — references job_postings.posting_id
skill_id	INTEGER (FK) — references skills.skill_id

3.6 Table: skill_trends (derived/aggregated)

Column	Type	Description
skill_id	INTEGER (FK)	References skills.skill_id
month	DATE	First day of the aggregation month
demand_count	INTEGER	Number of postings mentioning this skill that month
demand_share	FLOAT	demand_count normalized by total postings that month
avg_salary	FLOAT	Average salary midpoint across postings mentioning this skill
trend_class	ENUM (nullable)	Rising / Stable / Declining — populated by the Section 2.2 model

3.7 Table: layoffs

Column	Type	Description
layoff_id	INTEGER (PK)	Unique identifier
company_id	INTEGER (FK)	References companies.company_id
date	DATE	Date of reported layoff event
count	INTEGER (nullable)	Number of employees affected, if reported
percent_of_workforce	FLOAT (nullable)	Percentage of workforce affected, if reported
department	VARCHAR(100) (nullable)	Affected department, where source data provides it
source	VARCHAR(50)	Provenance, e.g. “layoffs.fyi”

3.8 Table: user_queries (application logging)

This table is not training data — it logs how the finished application is actually used, for the report's evaluation section and for future iteration.

Column	Type	Description
query_id	INTEGER (PK)	Unique identifier
input_skills	TEXT	Skills submitted by the user, as skill_id list
dream_company_id	INTEGER (FK, nullable)	May be null if company not in database
target_salary	FLOAT	User-stated salary expectation
timestamp	TIMESTAMP	Query time

3.9 Data Sources

- Job postings: publicly available structured datasets (e.g. Kaggle LinkedIn/Indeed job-postings collections), used in place of direct scraping to avoid Terms-of-Service issues.
- Skill trend enrichment: Stack Overflow Developer Survey (annual), used to extend and validate trend signal beyond the postings window.
- Layoff data: layoffs.fyi historical dataset (company, date, headcount, and where available, department).

4. Existing Systems and Comparative Analysis

This section surveys the closest existing tools in the AI-career-guidance space, evaluated specifically against the four capabilities CareerCompass targets: company-specific market-trend benchmarking, salary-expectation reframing against real data, skill-gap ranking grounded in trend data, and layoff-pattern awareness. It concludes with a direct comparison to general-purpose LLM assistants (Claude, ChatGPT, etc.), since students can and do already turn to these for career advice.

4.1 Surveyed Systems

4.1.1 Jenova AI — Career Advisor

Positions itself as a strategic career agent that analyzes a user's experience and goals against “market realities,” going beyond personality-quiz-style tools to identify specific skill gaps and estimated timelines to close them.

Drawbacks relative to CareerCompass: Guidance is generated conversationally per query rather than computed from a fixed, inspectable dataset; there is no visible mechanism for company-specific hiring-trend benchmarking or layoff-pattern context, and salary guidance is not tied to a transparent regression model a user or evaluator could audit.

4.1.2 Predictive Job-Matching Platforms (e.g. Eightfold.ai-class tools)

This category of tool uses ML, NLP, and neural networks to model employment trajectories, analyzing resumes, skills, behaviors, and market trends to forecast job fit, salary growth, and career transitions, claiming high workflow-matching accuracy on internal benchmarks.

Drawbacks relative to CareerCompass: These are enterprise-oriented platforms (typically sold to employers for talent acquisition, not directly to students), with closed, proprietary models and no free student-facing salary-expectation-vs-reality tool; layoff-context and skill-drop guidance are not part of their stated feature set.

4.1.3 Resume/Application-Focused Tools (Careerflow, Teal, Jobscan, Kickresume, Rezi)

This cluster of tools focuses on resume optimization, ATS keyword matching, and application tracking rather than market-trend prediction.

Drawbacks relative to CareerCompass: These tools address a different problem — application quality, not market-fit decision-making. They do not attempt company-trend benchmarking, salary prediction, or skill-gap-versus-market-demand analysis at all.

4.1.4 Enterprise Upskilling Platforms (Disco, Pluralsight, LinkedIn Learning, Gloat, CoachHub)

These integrate AI to surface industry trends and skill requirements as part of structured employee career-development programs within organizations.

Drawbacks relative to CareerCompass: Built for employed professionals inside a sponsoring organization, not for students evaluating external companies; no dream-company or layoff-aware benchmarking; typically paid enterprise licenses inaccessible to individual students.

4.1.5 General Job Platforms Adding AI Coaching (LinkedIn and peers)

Industry commentary indicates AI-powered career coaching is moving from a premium add-on to a standard, commoditized feature expected across major job platforms within the next few years.

Drawbacks relative to CareerCompass: Feature is still emerging and generic across all users of the platform; not tailored to a specific student's stated skill set, dream company, and salary expectation as three explicit joint inputs; underlying models are not disclosed or reproducible for academic evaluation.

4.2 Comparison Table

System	Company-specific trend benchmarking	Salary prediction transparency	Layoff-aware context	Explainable / reproducible model
Jenova AI Career Advisor	No	Low — conversational, not model-based	No	No — closed system
Enterprise predictive job-matching (Eightfold-class)	Partial (employer-side)	Low — proprietary	No	No — proprietary
Resume tools (Careerflow, Teal, etc.)	No	No	No	N/A — different problem scope
Enterprise upskilling platforms	No (internal-org focus)	No	No	No — closed system
General job platforms w/ AI coaching	Generic, not personalized to 3 joint inputs	Low	No	No — undisclosed models
General-purpose LLMs (Claude, ChatGPT)	Partial — only if searched live, not grounded in a fixed dataset	Low — estimated from training/search, not computed	Partial — only if explicitly asked and searched	No — not auditable or reproducible
CareerCompass (proposed)	Yes — from structured posting time-series	Yes — regression model with feature importance	Yes — dedicated layoffs table, correlational not causal	Yes — fully inspectable pipeline

4.3 Can Students Just Use an LLM (Claude, ChatGPT, etc.) Instead?

This question deserves a direct answer rather than being sidestepped, since it is the most likely challenge a student project of this kind will face in evaluation.

What general-purpose LLMs already do well: general “what skills should I learn for X role” advice drawn from broad training knowledge and, where web search is available, individual current data points such as a specific company's recent layoff announcement.

What they do not structurally provide, and what this project targets instead:

- No persistent, structured dataset the answer is grounded in — an LLM's answer about whether a company's hiring is growing or shrinking is pattern-matched from scattered text, not queried from a maintained time-series of postings.
- No quantitative trend modeling — a claim that a skill's demand is declining is a qualitative impression from text seen previously, not a computed regression slope over a defined historical window.
- No joint, consistent scoring of three simultaneous inputs (skills, dream company, salary expectation) against a numeric database — an LLM improvises a fresh answer per query with no guarantee of consistency across repeated queries or as underlying data updates.

- No reproducibility or auditability — a model's feature weights or a similarity score can be inspected and defended; an LLM's reasoning for a specific recommendation cannot be verified against source data in the same way.

The honest positioning, and the one this report recommends carrying into the project's introduction and viva defense, is that general-purpose LLMs are strong generalist advisors, while CareerCompass is a narrower, grounded, numeric, and updatable prediction system — a complementary tool rather than a like-for-like replacement.

5. Summary

CareerCompass is scoped as a system that combines genuine, justified machine learning (salary regression, skill-trend classification, similarity-based recommendation) with clearly labeled deterministic components (skill-gap ranking, trend aggregation), built on a purpose-designed relational schema sourced from publicly available job-postings, developer-survey, and layoff datasets. Its differentiation from both commercial AI-career tools and general-purpose LLM assistants rests on three properties none of the surveyed alternatives combine: company-specific trend benchmarking grounded in a maintained dataset, a transparent and auditable salary-prediction model, and layoff-pattern context presented honestly as correlational rather than causal.

This document is intended as the ideation and design reference for the implementation phase of the project; Section 2 model choices and Section 3 schema are written to be directly actionable as the starting point for the codebase.